Competitive Programming Lab – II
(Common to ECE, EEE, MECH and CIVIL**)**

Course Objectives:

- Understanding importance of Mathematics and Problem solving approaches for Programming
- Understanding importance of optimized solution for problem solving

Course Outcomes:

After completion of the course, the students will be able to

1. Analyze the problem using Mathematics
2. Select appropriate Sorting /Searching technique to solve the Problem
3. Develop solutions using stack and queue
4. Develop solutions using Non-Linear Data Structures
5. Develop Solution to problems using Dynamic Programming.

Lab Module – I

Number Theory

Class Test:	https://www.geeksforgeeks.org/batch/cp-mathematics/track/cp-mathprimeFactorization/problem/class-test
Largest prime factor:	https://www.geeksforgeeks.org/problems/largest-prime-factor2601/1
Integer to Roman:	https://leetcode.com/problems/integer-to-roman/description/
Total Chocolates II:	https://www.geeksforgeeks.org/batch/cp-mathematics/track/cp-math-primeFactorization/problem/total-chocolates-ii
Count Primes:	https://leetcode.com/problems/count-primes/description/
Make them Co-prime:	https://www.geeksforgeeks.org/batch/cp-mathematics/track/cp-math-primeFactorization/problem/make-them-co-prime

Lab Module – II

Number Theory

Number of 1 Bits:	https://www.geeksforgeeks.org/problems/set-bits0143/1
Bit Difference:	https://www.geeksforgeeks.org/problems/bit-difference-1587115620/1
Power of 2:	https://www.geeksforgeeks.org/problems/power-of-2-1587115620/1
Find position of set bit:	https://www.geeksforgeeks.org/problems/find-position-of-set-bit3706/1
Missing Number:	https://leetcode.com/problems/missing-number/description/
Non Repeating Numbers:	https://www.geeksforgeeks.org/problems/finding-the-numbers0215/1

Lab Module – III

Searching: Linear and Binary Search

Find First and Last Position of Element in Sorted Array:	https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/description/
Search in Rotated Sorted Array:	https://leetcode.com/problems/search-in-rotated-sorted-array/description/
Find Peak Element:	https://leetcode.com/problems/find-peak-element/description/
Single Element in a Sorted Array:	https://leetcode.com/problems/single-element-in-a-sorted-array/description/

Search a 2D Matrix:	https://leetcode.com/problems/search-a-2d-matrix/description/
Capacity to Ship Packages Within D Days:	https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/description/

Lab Module – IV

Sorting: Bubble Sort, insertion Sort, Quick Sort, Merge Sort, Counting Sort

Minimum Moves to Equal Array Elements:	https://leetcode.com/problems/minimum-moves-to-equal-array-elements/description/
Minimum Absolute Difference:	https://leetcode.com/problems/minimum-absolute-difference/description/
Sort Array By Parity:	https://leetcode.com/problems/sort-array-by-parity/description/
Max Chunks To Make Sorted:	https://leetcode.com/problems/max-chunks-to-make-sorted/description/
K^{th} Largest Element in an Array:	https://leetcode.com/problems/kth-largest-element-in-an-array/description/
Boats to Save People:	https://leetcode.com/problems/boats-to-save-people/description/?envType=problem-list-v2&envId=sorting

Lab Module – V

Recursion and Backtracking

Subsets	https://leetcode.com/problems/subsets/description/
Permutations	https://leetcode.com/problems/permutations/description/
Permutations II	https://leetcode.com/problems/permutations-ii/description/
Subsets II	https://leetcode.com/problems/subsets-ii/description/
Palindrome Partitioning	https://leetcode.com/problems/palindrome-partitioning/description/
N-Queens	https://leetcode.com/problems/n-queens/description/

Lab Module – VI

Linked List part1: Singly Liked List, Doubly Linked List, Circular Linked List

Middle of the Linked List	https://leetcode.com/problems/middle-of-the-linked-list/description/
Delete Node in a Linked List	https://leetcode.com/problems/delete-node-in-a-linked-list/description/
Reverse Linked List:	https://leetcode.com/problems/reverse-linked-list/description/
Convert Binary Number in a Linked List to Integer:	https://leetcode.com/problems/convert-binary-number-in-a-linked-list-to-integer/description/
Palindrome Linked List	https://leetcode.com/problems/palindrome-linked-list/description/
Remove Linked List Elements	https://leetcode.com/problems/remove-linked-list-elements/description/

Lab Module – VII

Linked List part2:

Sort List:	https://leetcode.com/problems/sort-list/description/
Remove Duplicates from Sorted List:	https://leetcode.com/problems/remove-duplicates-from-sorted-list/description/
Merge Two Sorted Lists:	https://leetcode.com/problems/merge-two-sorted-lists/description/
Intersection of Two Linked	https://leetcode.com/problems/intersection-of-two-linked-

Lists:	lists/description/
Linked List Cycle:	https://leetcode.com/problems/linked-list-cycle/description/
Linked List Cycle II:	https://leetcode.com/problems/linked-list-cycle-ii/description/

Lab Module – VIII

Stack & Queue

Evaluate Reverse Polish Notation:	https://leetcode.com/problems/evaluate-reverse-polish-notation/description/
Min Stack:	https://leetcode.com/problems/min-stack/description/
Daily Temperatures:	https://leetcode.com/problems/daily-temperatures/description/
Largest Rectangle in Histogram:	https://leetcode.com/problems/largest-rectangle-in-histogram/description/
Implement Stack using Queues:	https://leetcode.com/problems/implement-stack-using-queues/description/
Implement Queue using Stacks:	https://leetcode.com/problems/implement-queue-using-stacks/description/
Gas Station:	https://leetcode.com/problems/gas-station/description/
Sliding Window Maximum:	https://leetcode.com/problems/sliding-window-maximum/description/

Lab Module – IX

Hashing

Find Common Elements Between Two Arrays	https://leetcode.com/problems/find-common-elements-between-two-arrays/description/
Contains Duplicate	https://leetcode.com/problems/contains-duplicate/description/
Find All Duplicates in an Array	https://leetcode.com/problems/find-all-duplicates-in-an-array/description/
Sort Characters By Frequency	https://leetcode.com/problems/sort-characters-by-frequency/description/
Group Anagrams	https://leetcode.com/problems/group-anagrams/description/
Isomorphic Strings	https://leetcode.com/problems/isomorphic-strings/description/

Lab Module – X

DP: Linear DP,2 Dimensional DP, DP on Grids,

Climbing Stairs:	https://leetcode.com/problems/climbing-stairs/description/
Decode Ways:	https://leetcode.com/problems/decode-ways/description/
House Robber:	https://leetcode.com/problems/house-robber/description/
Partition Equal Subset Sum:	https://leetcode.com/problems/partition-equal-subset-sum/description/
Frog Jump:	https://leetcode.com/problems/frog-jump/description/
Unique Paths:	https://leetcode.com/problems/unique-paths/description/
Minimum Path Sum:	https://leetcode.com/problems/minimum-path-sum/description/
Knapsack 1:	https://atcoder.jp/contests/dp/tasks/dp_d
Longest Increasing Subsequence:	https://leetcode.com/problems/longest-increasing-subsequence/description/
Largest Divisible Subset:	https://leetcode.com/problems/largest-divisible-subset/description/

Lab Module – XI

Tree:

Binary Tree Preorder Traversal:	https://leetcode.com/problems/binary-tree-preorder-traversal/description/
Binary Tree Inorder Traversal	https://leetcode.com/problems/binary-tree-inorder-traversal/description/
Binary Tree Postorder Traversal	https://leetcode.com/problems/binary-tree-postorder-traversal/description/
Maximum Depth of Binary Tree	https://leetcode.com/problems/maximum-depth-of-binary-tree/description/
Diameter of Binary Tree	https://leetcode.com/problems/diameter-of-binary-tree/description/
Binary Tree Right Side View	https://leetcode.com/problems/binary-tree-right-side-view/description/
Path Sum	https://leetcode.com/problems/path-sum/description/

Lab Module – XII

Graph: DFS and BFS on Graphs, Topological Sort, Flood Fill

DFS Traversal:	https://www.naukri.com/code360/problems/dfs-traversal_630462
Find if Path Exists in Graph:	https://leetcode.com/problems/find-if-path-exists-in-graph/description/
Keys and Rooms:	https://leetcode.com/problems/keys-and-rooms/description/
Open the Lock:	https://leetcode.com/problems/open-the-lock/description/
Jump Game III:	https://leetcode.com/problems/jump-game-iii/description/
Course Schedule:	https://leetcode.com/problems/course-schedule/description/

Text Books:

1. T.H. Cormen et.al. – Introduction to Algorithms – PHI, New Delhi, 2005.
2. E. Horowitz. et.al., Fundamentals of computer Algorithms, Universities Press, 2008, 2nd Edition.

Reference Books:

1. G.Brassard & P. Bratley – Fundamentals of Algorithms, PHI, New Delhi, 2005.
2. S.Dasgupta et.al. – Algorithms, TMH, New Delhi – 2007.
3. Silberschatz, Korth :Database System Concepts. McGraw hill,5th Edition.
4. Elmasri, Navrate:Fundamentals of Database Systems. Pearson Education,6th Edition

Reference Links

1. https://leetcode.com/problems/
2. https://nptel.ac.in/courses/106106131/
3. https://www.spoj.com/problems